

Area of Shapes — Answer Key

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Part A — Triangles · $A = \frac{bh}{2}$

1) $b=10, h=4. \frac{10 \cdot 4}{2} = 20$ **20 cm²**

2) $b=8, h=5. \frac{8 \cdot 5}{2} = 20$ **20 m²**

3) $b=12, h=7. \frac{12 \cdot 7}{2} = 42$ **42 in²**

4) $b=9, h=6. \frac{9 \cdot 6}{2} = 27$ **27 yd²**

5) $b=15, h=4. \frac{15 \cdot 4}{2} = 30$ **30 km²**

6) $b=8, h=5.2$ (ignore 6 slant). $\frac{8 \cdot 5.2}{2} = 20.8$
20.8 in²

7) $b=11.6, h=6. \frac{11.6 \cdot 6}{2} = 34.8$ **34.8 km²**

Part B — Parallelograms · $A = bh$

8) $9 \cdot 6 = 54$ **54 cm²**

9) $12 \cdot 8 = 96$ **96 km²**

10) $14 \cdot 5 = 70$ **70 m²**

11) $9.7 \cdot 5 = 48.5$ (ignore 5.8 slant) **48.5 yd²**

Part C — Trapezoids · $A = \frac{(b_1 + b_2)h}{2}$

★ 12) $b_1=3$, $b_2=6$, $h=4$ (cm) — shown BOTH ways

WAY A — Two trapezoids → parallelogram → half

Parallelogram base = $3 + 6 = 9$, height = 4.

Parallelogram area = $9 \times 4 = 36$

Trapezoid = half = $\frac{36}{2} = 18$

WAY B — Diagonal split → sum of two triangles

Triangle 1 (top): $\frac{3 \times 4}{2} = 6$

Triangle 2 (bottom): $\frac{6 \times 4}{2} = 12$

Sum = $6 + 12 = 18$

18 cm²

Both ways → same answer ✓

13) $\frac{(5 + 11) \cdot 4}{2} = \frac{64}{2} = 32$ **32 m²**

14) $\frac{(4 + 8) \cdot 3}{2} = \frac{36}{2} = 18$ **18 km²**

15) $\frac{(4 + 1) \cdot 3.2}{2} = \frac{16}{2} = 8$ **8 yd²**

16) $\frac{(6 + 3) \cdot 3.9}{2} = \frac{35.1}{2} = 17.55$ **17.55 m²**

17) $\frac{(4.6 + 11.6) \cdot 6}{2} = \frac{16.2 \cdot 6}{2} = \frac{97.2}{2} = 48.6$

48.6 km²

Part D — Reverse the formula

18) Triangle area 24, $h=6$. $24 = \frac{b \cdot 6}{2} = 3b \rightarrow b = 8$

b = 8 cm

19) Parallelogram area 72, $b=9$. $72 = 9h \rightarrow h = 8$

h = 8 m

20) Trapezoid area 96, $h=8$, $b_1=10$. $96 = \frac{(10 + b_2) \cdot 8}{2} = 4(10 + b_2) \rightarrow 10 + b_2 = 24 \rightarrow b_2 = 14$ **b₂ = 14 cm**